Feedback

Tomasz Włodarczyk

2025-08-07

Feedback as a self-regulation mechanism in systems, involving the transmission of information about the outcome of actions, enabling correction, adaptation, and optimization of processes.

Introduction:

Feedback is a key communication mechanism in cybernetics theory. It is a process in which the result of a system's action influences its further functioning, creating a mechanism for self-regulation and adaptation.

Characteristics:

Feedback is an informational mechanism that allows systems to:

1. Automatically correct their actions
2. Adapt to changing conditions
3. Maintain balance
4. Optimize processes

Types of Feedback:

- **Positive (reinforcing)** - amplifying changes in the system
- **Negative (damping)** - stabilizing and controlling processes

Practical Examples from Various Fields:

Psychology and Social Relationships:

1. Conversations where participants mutually adjust their communication
2. Psychological therapy - analyzing patient behaviors
3. Parenting processes - parental responses to a child's behavior

Economics and Business:

1. Customer satisfaction surveys
2. Employee reports and performance evaluations
3. Market mechanisms correcting product prices
4. Motivational systems in companies

Technology:

1. Thermostats regulating temperature
2. GPS navigation systems
3. Automatic control in robotics
4. Machine learning algorithms

Biology:

1. Regulation of body temperature
2. Immune system mechanisms
3. Metabolic processes
4. Homeostasis

Communication:

1. Mediation and negotiations
2. Group discussions
3. Democratic processes
4. Team communication

Conclusions:

Feedback is a universal adaptive mechanism present in all complex systems. It enables self-regulation, improvement, and effective functioning through continuous information exchange and action correction.

More

O autorze Fr. Dr. Tomasz Włodarczyk

© 2026 Fr. Dr. Tomasz Włodarczyk

Obraz Robert A. Taylor, U.S. Department of Energy - “Feedback”. In: U.S. Department of Energy’s Introduction to System Dynamics.Public domain via Wikipedia Commons